

BLP.

$$\hat{Y} = \hat{\beta}^T X \rightarrow \text{prediction.}$$

$$E(Y - \hat{Y})^2 = 0$$

$$s^2 = E(Y - \hat{\beta}^T X)^2 \rightarrow \text{BLP}$$

$$\frac{\partial s}{\partial \beta} \bigg|_{\beta = \hat{\beta}} = 0$$

$$\sigma_{y_n} = \sum_n \hat{\beta}$$

$$\hat{\beta} = \sum_n^{-1} \sigma_{y_n}$$

$$Q: \text{var}(Y - \hat{\beta}^T X)$$

$$\text{var}(Y) + \text{var}(\hat{\beta}^T X) - 2 \text{cov}(Y, \hat{\beta}^T X)$$

$$\Rightarrow \sigma_{yy} + \hat{\beta}^T \hat{\beta} - 2 \hat{\beta}^T \text{cov}(Y, X)$$

$$\sigma_{yy} + \sigma_{y_n}^T \Sigma_n^{-1} \sigma_{y_n} - 2 \sigma_{y_n}^T \Sigma_n^{-1} \sigma_{y_n}$$

$$= (\sigma_{yy} - \sigma_{y_n}^T \Sigma_n^{-1} \sigma_{y_n})$$

Q. Prove: $\hat{\beta}$ is unique.

Now, let, $\hat{\beta}_0, \hat{\beta}$

$$E(\hat{\beta}_0^T X - \hat{\beta}^T X) = 0$$

$$\Rightarrow E[(\hat{\beta}_0^T - \hat{\beta}^T) X] = 0$$

$$(\hat{\beta}_0 - \hat{\beta})^T \sum_x (\hat{\beta}_0 - \hat{\beta}) = 0$$

$$\sum_x \hat{\beta}_0 = \sigma_{yx}^T, \quad \sum_x \hat{\beta} = \sigma_{yx}$$

$$P[\hat{\beta}_0 = \hat{\beta}] = 1$$

Q. Corr $(Y, \hat{\beta}^T X)$

$$\frac{[Cov(Y, \hat{\beta}^T X)]^2}{Var(Y) Var(\hat{\beta}^T X)}$$

$$\frac{(\sigma_{yx} \hat{\beta})^2}{\sigma_{yy} (\hat{\beta}^T \sum_x \hat{\beta})}$$

$$\frac{Cov(Y, \hat{\beta}^T X)}{\sigma_{yx} \hat{\beta}}$$

$$\Rightarrow \frac{\sigma_{yx}^T \sum_x \sigma_{yx}}{\sigma_{yy}}$$

$$\underline{\underline{Q. \text{ cov}[(Y - \hat{\beta}^T X), x_j] = 0}}$$

$$Q_j \text{ } e_j = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad e_j^T X$$

$$\text{cov}(Y - \hat{\beta}^T X, e_j^T X) \quad \begin{matrix} \text{var}(\hat{\beta}^T X) \\ \text{cov}(\hat{\beta}^T X, \hat{\beta}^T X) \\ \hat{\beta}^T \underline{\underline{\text{cov}(X, X)}} \hat{\beta} \end{matrix}$$

$$\text{cov}(Y, e_j^T X) - \text{cov}(\hat{\beta}^T X, e_j^T X)$$

$$\frac{\text{cov}(Y, X) e_j}{e_j^T \sigma_{y|x} e_j} - \hat{\beta}^T \text{cov}(X, X) e_j$$

$$= \frac{(\sum x \sigma_{y|x})^T}{\sum x} e_j$$

$$= \sigma_{y|x}^T e_j$$

$$\Rightarrow 0$$